PROJECT STATUS / COLLABORATION BRIEF

MEHLISSA Next

A reproducible multilayer research platform for body transport, molecular interaction, cell response, and Nano-IoT communication

M0-M7	PASSED	CURRENT FOCUS
Architecture and implementation	All milestone gates	UX-2 CI verification, then UX-3 reporting

Prepared for prospective contributors and research partners
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Branch: mehlissa-next-generation | Base revision: a96b4ce
Repository: github.com/RegineWendt/MEHLISSA

Executive summary

MEHLISSA Next has progressed from a set of valuable historical prototypes to a coherent, modular research simulation platform. The project now implements the complete architectural path envisioned for the first dissertation-driven demonstrator: systemic transport, a replaceable lung model, capillary transport and molecular channels, receptor and intracellular cell response, nanodevice communication, an active gateway, a body area network, and external analysis.

Key achievement: All major layers can now participate in one reproducible FP9/lung fingerprinting workflow while remaining independently replaceable behind explicit contracts.

- Milestone gates M0 through M7 have passed their documented acceptance reviews.
- The implementation uses C++20, CMake, vcpkg, strict JSON Schemas, typed SI

quantities, stable error contracts, deterministic random streams, and explicit provenance.

- The base revision passes all 280 local Windows/MSVC tests and GitHub

Windows/MSVC, Linux/GCC, and Linux/Clang CI. The Clang path also passes formatting, clang-tidy, AddressSanitizer, and UndefinedBehaviorSanitizer.

- The UX-2 candidate adds a validated catalog of five model families and ten

curated starter configurations. All 281 local Windows/MSVC tests pass; cross-platform CI is the remaining acceptance check.

- The English User Guide, software architecture guide, model documentation,

traceability matrix, and architecture decision records support international collaboration.

- The result is a reproducible research-software demonstrator. It is not a

clinical assay model, a medical device, or a patient-specific digital twin.

Dimension	Assessment
Software architecture	Strong modular foundation with explicit ownership, conservation, lifecycle, configuration, and evidence boundaries.
End-to-end integration	Complete first software vertical slice through fingerprinting Levels A-E.
Scientific validation	Mixed maturity: verified equations, literature-parameterized candidates, selected independent comparisons, and multiple synthetic mechanisms.
User experience	UX-1 exposes the complete M7 demonstrator; UX-2 locally adds model/example discovery and safe starter copying. Rich result reports, campaigns, Python access, and a workbench remain.
Clinical readiness	Not claimed. Patient prediction, diagnosis, and treatment recommendations are explicitly outside the present scope.

How to read project labels

Short identifiers preserve traceability across requirements, code, tests, and evidence. They are not assumed knowledge in this report.

Label	Plain-language meaning
M0 through M8	Sequential milestone gates. M7 is the accepted complete fingerprinting software demonstrator. The future M8 gate concerns a governed research digital twin.
M7.4	A numbered increment within a milestone. M7.4 adds concentration- and receptor-binding-based fingerprint detection.
Level A	Historical-timing baseline for the fingerprinting scenario; it reproduces published dissertation event semantics and is not a separate “A run”.
Level B	Mechanistic fingerprint detection based on concentration, receptor binding, and a threshold.
Level C	Explicit release and complete/incomplete assembly of nine fingerprint tiles.
Level D	Executed local, gateway, body-area-network, and external-station communication.
Level E	Sensitivity, specificity, false-positive/false-negative, robustness, and misclassification analysis.
P0 through P3	Roadmap priority tiers: indispensable foundation, dissertation core, research-platform expansion, and long-term vision. They are not completion states.
UX-1 through UX-6	Ordered post-M7 user-experience packages, from one-command execution to a graphical research workbench.
run	One reproducible execution of an experiment or scenario. A qualified term such as “baseline run” must state what makes that execution distinct.

FP9 is the nine-part molecular fingerprint used by the first complete scenario. BVS is the historical BloodVoyagerS whole-body distribution reference. BAN is the body area network between gateway and external station. ODE denotes an ordinary differential-equation model, and SSA denotes a stochastic simulation algorithm.

1. Conceptual system architecture

MEHLISSA separates the research question from the models used to answer it. A scenario selects versioned model cards, initializes independent components, coordinates explicit transfers and events, and produces bounded results with provenance. This permits a fast surrogate, population model, field model, or detailed particle model to be compared without embedding medical special cases in the kernel.

The implemented conceptual path is:

1. **Body:** systemic circulation and whole-body transport.
2. **Lung or another organ:** replaceable organ-specific models.
3. **Capillary bed:** local transit, exchange, entity fate, and molecular channels.
4. **Cell:** receptor binding, intracellular signaling, delivery, and response.
5. **Nano-IoT:** nanodevice endpoints, local links, bounded relays, and gateway.
6. **BAN and station:** external measurement and a governed return command.

Binding engineering principles are:

- **Layer independence:** body, organ, capillary, cell, and communication models live in separate libraries.
- **Replaceable resolution:** model variants share stable boundaries while documenting their own validity scopes.
- **Explicit ownership and conservation:** identities and physical amounts cannot silently disappear or be duplicated at a boundary.
- **Dimension-safe internals:** public numerical APIs use typed quantities; files state units explicitly.
- **Reproducibility:** runs bind time, seed, named random streams, model versions, hashes, logs, and checkpoints.
- **Strict configuration:** versioned JSON Schemas and semantic loaders reject unknown or inconsistent inputs.
- **Evidence before claims:** verification, calibration, independent validation, historical regression, and sensitivity analysis remain distinct.
- **Bounded scale:** populations, aggregates, and record limits prevent uncontrolled output growth.

Independent MEHLISSA Next code is licensed under MPL-2.0. Legacy code and direct ports remain GPL-2.0-only. New original documentation and approved original data are licensed under CC BY 4.0. Artifact-specific provenance and license sidecars prevent unclear data rights from entering public packages.

2. Milestones achieved: M0-M3

Gate	Plain-language result	Representative capabilities
M0	Project charter and architecture decisions	Dissertation requirements and traceability; new-kernel decision; four-layer architecture; lung selected as first organ; licensing, data inventory, users, workflows, validation gaps, and partner roles.
M1	Trustworthy simulation kernel	Monotonic nanosecond time; 3D position; typed SI units; named deterministic random streams; component lifecycle and ownership; experiment manifests; provenance; structured errors and JSONL logs; checkpoints; byte-stable cross-platform reference.
M2	Validated body transport layer	Schema-validated 95-segment vascular graph; approved legacy migration; graph and vessel-9 branching invariants; injection, extraction, and identity-preserving transport; bounded observations; rest, exercise, and posture overlays; 6,359/63,590-agent BVS regression.
M3	Body-organ coupling and lung reference	Typed body-organ transfers; coarse, serial-region, pulmonary 0D, flow-adaptive, age-conditioned, pressure-distensible, and five-lobe parallel-bed models; source-disjoint validation paths; same-scenario resolution comparison; historical FP9 timer baseline.

M0-M3 establish that a medical scenario no longer needs its own simulation engine or vascular implementation. The kernel remains generic, a vascular graph can change without recompilation, and a caller can select a lung model at a declared resolution while preserving the meaning and ownership of transported entities and quantities.

The pulmonary models include literature-parameterized and independently evaluated endpoints, but remain zero-dimensional or equivalent representations. They do not yet provide anatomical one-dimensional flow, pulsatile pulmonary mechanics, patient imaging, or a complete jointly measured validation cohort.

3. Milestones achieved: M4-M7

Gate	Plain-language result	Representative capabilities
M4	Capillary communication layer	Executable arteriole-capillary-venule route; geometry and continuity closure; dynamic recruitment; balanced blood/endothelium/interstitium/cell exchange; residence and terminal disposition; analytical diffusion, Brownian endpoint, trajectory, radial finite-volume, and shared axial advection-reaction models.
M5	Cell response layer	Analytical and time-varying receptor-ligand binding; stochastic finite-receptor SSA; capillary-to-cell signal hand-off; shared intracellular ODE/SSA network; conservative device release and uptake; synthetic apoptosis event; cohort-compressed populations up to one trillion represented cells.
M6	End-to-end Nano-IoT plane	Nanodevices and local messages; one-hop delivery outcomes; bounded routing and relays; active gateway uplink/downlink; BAN and governed station loop; payload-free external network-simulator adapter; twelve fail-closed resilience and boundary-misuse cases.
M7	Holistic fingerprinting vertical slice	Strict selection of 13 scenario artifacts; ten-stage causal contract; typed runtime; identity trace; artifact SHA-256 hashes; concentration-driven binding and negative control; nine FP9 tiles and incomplete control; locator-to-station communication; Level-E classification metrics and Wilson intervals; deterministic result schema 2.0.0.

In one sentence: MEHLISSA can now execute a reproducible software workflow in which a configured multilayer stack produces a biological detection event, assembles an explicit fingerprint, communicates it through local and external boundaries, and reports the outcome with complete artifact identity and declared uncertainty limitations.

Positive, negative, invariant, numerical, integration, regression, and schema tests cover all accepted milestones. All compiler and analysis CI jobs pass. Gate reviews explicitly prevent a software pass from being presented as physiological or clinical validation. The English User Guide is mandatory review evidence at every future milestone and UX-package review.

4. How MEHLISSA can be used today

Access route	Suitable uses	Current examples
Command line	Direct use without writing C++	Validate manifests and vascular graphs; run the minimal workflow and BVS regression; apply body-state overlays; validate and execute the complete M7 demonstrator.
Reference and benchmark drivers	Reproduce accepted scientific and numerical comparisons	Pulmonary validations; coarse-versus-five-lobe comparison; molecular-channel comparisons; historical FP9 timing; benchmark campaigns.
C++ component APIs	Compose or extend advanced research workflows	Organ-capillary round trip; cell and communication models; external simulator adapter; direct access to the same M7 runtime used by the CLI.
Automated test suite	Audit contracts and reproduce gate evidence	Focused M1-M7 CTest filters, negative controls, deterministic replay, conservation, and identity checks.

Representative research questions include:

- How does injection site or physiological state affect systemic distribution and transit?

- Which outputs change when a coarse lung surrogate is replaced by five parallel pulmonary lobe beds?

- How do binding, intracellular signaling, device release, uptake, and a synthetic cell response connect without violating amount ownership?

- How do loss, corruption, expiry, replay, routing errors, or station-policy rejection affect a Nano-IoT measurement path?

- Can the complete FP9/lung workflow preserve causal identity, report incomplete fingerprints, and compute classification metrics reproducibly?

The complete M7 scenario can now be discovered, validated, executed, and summarized through the normal application:

```
build/windows-msvc/apps/Debug/mehlissa.exe scenario list
build/windows-msvc/apps/Debug/mehlissa.exe scenario run --file
examples/scenarios/fp9-lung-level-a-v1.json --output results/fp9-reference
```

The separate `scenario validate` command checks the same inputs without executing them. The output argument is a parent directory; each run creates a unique UTC-labelled child containing `result.json`, `provenance.json`, `run.log.jsonl`, and `summary.txt`; the application prints the exact path. An existing result can be summarized with `mehlissa result summarize --file <result.json>`. The command reuses the accepted M7 composer and holistic runner, validates all thirteen selected definition/schema pairs before execution, and preserves the non-clinical interpretation boundary.

UX-2 adds discovery without creating a second source of scientific truth:

```
mehlissa model list
mehlissa model describe --id organ.pulmonary-zero-dimensional
mehlissa example list --model organ.pulmonary-zero-dimensional
mehlissa example copy --id scenario.fp9-complete --output work/fp9-study
```

The strict versioned catalog describes five model families and ten licensed starters, including maturity, validity, evidence, limitations, key parameter paths, artifacts, and documentation. Checks reject duplicate IDs, unknown model references, missing assets, and paths outside the repository. Copying preserves the license and refuses to overwrite work. All 281 local Windows/MSVC tests pass; cross-platform CI is the final acceptance step.

5. Current personalization capability

MEHLISSA supports parameterized research profiles, but not a complete virtual person. A contributor can derive a versioned model card, state whether values are measured, literature-derived, calibrated, assumed, or used only for sensitivity analysis, and preserve units, uncertainty, provenance, and limitations.

Scope	Available now	Not yet available
Body	Load any schema-valid vascular graph; apply compatible cardiac-output and transition overlays.	Production import of segmented personal anatomy; complete pressure/compliance and disease physiology.
Lung	Configure pressure boundary, resistance, compliance, flow, transit, age band, distensibility, and lobe shares.	Automatic parameter inference; geometry-resolved patient lung; complete personal validation.
Capillary	Configure equivalent geometry, flow, recruitment, exchange, observation, and disposition.	Patient microvascular anatomy, hematocrit/rheology, qualified kinetics, and spatial tissue coupling.
Cell	Configure channel, receptor, reaction, delivery, threshold, response, and cohort parameters.	Patient-specific biomarker calibration, mechanistic apoptosis, and disease-specific validated families.
Whole run	Fix seeds, versions, checksums, output paths, and interpretation limits.	Canonical patient manifest, automatic cross-layer composition, parameter estimation, uncertainty propagation, and longitudinal updates.

Identifiable patient data must not be committed to the repository. Clinical or participant-specific work requires institutional approval plus explicit consent, pseudonymization, retention, access-control, and provenance workflows. The future M8 Research Digital Twin gate still forbids clinical decision claims without an appropriate regulatory and validation path.

6. Scientific limitations after M7

Area	What is demonstrated	What remains unproven
Cross-layer timing	One deterministic causal workflow with historical timing evidence.	Anatomically resolved localization and collector-return timing predicted by the coupled models.
Fingerprint biology	Concentration-driven receptor detection and explicit target identity.	Qualified FP9 gene-product combination, disease-marker data, and biological parameter validation.
Tile assembly	Nine explicit tile identities, an all-required rule, and incomplete negative control.	Executed NetTAS or molecular self-assembly physics and independently validated assembly kinetics.
Communication	Replaceable links, routing, gateway, BAN, station, metrics, and external-simulator boundary.	Calibrated molecular, intrabody, wearable, Bluetooth, or IEEE 802.15.6 behavior.
Classification	Correct sensitivity/specificity accounting, labelled cases, and Wilson intervals.	Empirical prevalence, clinical sensitivity/specificity, patient variability, and diagnostic utility.
Physiology	Literature-scoped lung models and selected independent endpoint comparisons.	Jointly measured cohorts, broad disease states, pulsatility, anatomy, hematocrit, and full uncertainty propagation.
Therapeutic response	Conservative release/uptake and a bounded synthetic cell-response event.	Mechanistic pharmacology, spatial drug transport, metabolism, toxicity, and treatment efficacy.

Interpretation rule: A verified software contract shows that the declared mechanism is implemented consistently. It does not, by itself, show that the mechanism predicts a biological or clinical outcome.

7. Remaining development program

Program	Principal work	Intended outcome
Usability and orchestration	UX-1 provides the complete-scenario CLI and UX-2 locally provides model/example discovery and safe starter copying; richer reports, campaigns, Python access, and a workbench remain.	A researcher can find suitable models and starters and run M7 without C++ knowledge or test binaries, then progressively gain better analysis tools.
Scientific qualification	Replace historical or synthetic assumptions with measured parameters, calibration data, independent validation, uncertainty, and sensitivity campaigns.	Defensible scenario-specific conclusions within explicit scopes.
Additional medical scenarios	Continuous monitoring, liquid biopsy, endocrine/adrenal venous sampling, CAR-T, and ultimately metastasis prevention.	Evidence that the architecture generalizes beyond fingerprinting.
Additional organs	Begin with a kidney surrogate and progress toward regional filtration, clearance, and organ-specific capillary interfaces.	Evidence that organ factories and coupling contracts are not lung-specific.
Personalization and digital twin	Patient manifest, imaging and SimVascular/CFD import, parameter estimation, identifiability, longitudinal updates, and governance.	Research-twin maturity from geometry to physiology and biochemistry.
Scaling and HPC	Bound output, profile CPU/memory/I/O, use population and field representations, and parallelize safe work.	Large ensembles and sensitivity studies without sacrificing validated behavior.
Visualization and analysis	Standardized result readers, plots, run comparison, uncertainty views, and reproducible figure export.	Results that non-developers can inspect and communicate.

The roadmap priority tiers have the following meaning in the current state:

- **P0, indispensable foundation, is substantially complete:** kernel, reproducibility, CI, units, schemas, and the BVS baseline.
- **P1, dissertation core, is substantially complete at demonstrator level:** body, lung, capillary, cell, Nano-IoT, and fingerprinting vertical slice.
- **P2, platform expansion, is now the main scientific expansion space:** new scenarios, multiple organs, external models, and integrated visualization.
- **P3, long-term vision, remains open:** metastasis prevention, patient-specific models, a dynamic digital twin, HPC ensembles, and eventual clinical validation where appropriate.

8. Post-M7 usability program

UX means user experience. These packages improve access and interpretation; they do not by themselves increase physiological or clinical validity.

Package	Plain-language objective	Status
UX-1 - One-command M7 scenario execution	Validate and run the complete fingerprinting demonstrator and summarize its result through the normal application.	Passed; all 280 local Windows/MSVC tests and all supported GitHub CI jobs pass
UX-2 - Model and example discovery	List and describe available artifacts, parameters, evidence, and limitations.	Implemented locally; five model families, ten examples, and all 281 Windows/MSVC tests pass; cross-platform CI pending
UX-3 - Human-readable and HTML result reporting	Provide concise terminal, tabular, and shareable HTML views over the complete machine-readable result.	Planned
UX-4 - Derived experiments and campaigns	Create controlled variants, replicates, parameter sweeps, paired comparisons, and aggregate analyses.	Planned
UX-5 - Python API and notebooks	Support common scientific analysis and plotting workflows without replacing the C++ implementation authority.	Planned
UX-6 - Graphical research workbench	Add guided scenario editing, run control, comparison, provenance, and uncertainty visualization after the interfaces are stable.	Planned

The intended first user experience is:

```
mehlissa scenario list
mehlissa scenario validate --file examples/scenarios/fp9-lung-level-a-v1.json
mehlissa scenario run --file examples/scenarios/fp9-lung-level-a-v1.json --output
results/fp9-reference
mehlissa result summarize --file results/fp9-reference/<printed-run-directory>/result.json
```

UX-1 now reuses the existing M7 implementation, validates every selected artifact before execution, creates a unique run directory, writes validated result/provenance/log/summary artifacts, and reports stable actionable errors. Its automated end-to-end test covers help, listing, positive validation, negative validation, execution, artifact completeness, and re-summarization. The Windows MSVC build and all 280 local tests pass. GitHub CI run 33620386319 also passes Windows/MSVC, Linux/GCC, and Linux/Clang analysis and sanitizer jobs. UX-1 is therefore fully accepted and UX-2 is the active package.

The UX-2 candidate adds `model list`, `model describe`, filtered `example list`, and fail-safe `example copy`. The application validates the strict catalog and its repository references before displaying or copying anything. All 281 local Windows/MSVC tests pass; successful supported GitHub CI remains the final gate before UX-2 is marked accepted and UX-3 becomes active.

9. Opportunities for collaborators

Contributor profile	High-value starting point
C++ and simulation engineering	General scenario runner, multirate orchestration, model factories, profiling, and regression infrastructure.
Hemodynamics and pulmonary physiology	Review pulmonary cards, qualify cohorts, refine state dependence, and develop anatomical or pulsatile variants.
Renal physiology	Define and validate a kidney model family using the existing organ and capillary contracts.
Molecular communication	Calibrate local channels, implement assembly physics, and connect an external network simulator through the existing adapter.
Cell biology and pharmacology	Replace synthetic receptor, signaling, apoptosis, and delivery assumptions with evidence-qualified models.
Statistics and uncertainty quantification	Design independent validation, global sensitivity, uncertainty propagation, identifiability, and ensemble analysis.
Research UX and visualization	CLI discovery, result summaries, HTML reports, Python notebooks, and comparative visualization.
Research data and governance	Model-card provenance, licensing, FAIR data, patient-data controls, and digital-twin governance.

A new module is reviewable when its public contract and units are explicit; misuse fails predictably; time, identity, ownership, and conservation invariants pass; randomness is reproducible; output is bounded; schemas and examples agree; software verification is separated from calibration and validation; evidence and limitations are documented; and all supported CI paths pass.

10. Key project references

Document	Purpose
`docs/USER_GUIDE.md`	Non-technical introduction, experiment families, installation, commands, model workflows, interpretation, and troubleshooting.
`docs/architecture/SOFTWARE_ARCHITECTURE.md`	System structure, APIs, extension workflow, kidney outline, external simulator integration, and personalization.
`docs/ROADMAP.md`	Guiding principles, milestones, medical scenarios, UX packages, personalization, scaling, and cross-cutting programs.
`docs/requirements/SYSTEM_REQUIREMENTS.md`	Numbered requirements derived from the dissertation and project decisions.
`docs/requirements/TRACEABILITY_MATRIX.md`	Implementation and verification status for every tracked requirement.
`docs/m7/M7_GATE_REVIEW.md`	Formal decision and scientific limitations of the holistic fingerprinting demonstrator.
`docs/m0` **through** `docs/m6`	Gate reviews, model notes, validation evidence, and accepted limitations for preceding milestones.

- **Repository:** https://github.com/RegineWendt/MEHLISSA
- **Verified CI run:** https://github.com/RegineWendt/MEHLISSA/actions/runs/33620386319

MEHLISSA Next now provides the architectural and executable backbone required by the dissertation vision. The next phase is to make this backbone easier to run, broaden it to further scenarios and organs, and progressively replace synthetic assumptions with independently validated scientific models.